

Navdeep Dhull

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Summary:

AI Engineer and Machine Learning Engineer with 4+ years of experience building generative AI systems, LLM infrastructure, and machine learning pipelines. Experience in designing RAG architectures, agent frameworks, and low-latency inference systems on AWS. Proven track record of translating technical challenges into measurable cost savings and business growth

Professional Experience:

Stealth AI Startup- Sr. Gen AI Engineer,

May 2025 – Present

Architecting and scaling generative AI infrastructure, multi-agent frameworks, and search pipelines for enterprise deployment

- Reduced LLM inference latency by 45% for daily users by implementing vLLM for continuous batching and deploying Llama-3 models on AWS EC2, after identifying token bottleneck issues during peak traffic spikes
- Improved RAG retrieval precision by 28% by designing a hybrid search pipeline combining dense embeddings with sparse BM25 keyword scoring and a cross-encoder reranker
- Saved \$3,000 monthly in API expenditure by implementing semantic caching with Redis and routing low-complexity user queries to smaller, fine-tuned open-source models
- Automated compliance tracking for financial documents by building a multi-agent framework using LangGraph, allowing agents to extract, verify, and cross-reference structured tables from unstructured data

One Four Seven- Machine Learning Engineer,

Nov 2020 – Dec 2023

Led the development of personalized recommendation engines, data pipelines, and NLP models to optimize user engagement

- Improved click-through rate (CTR) by 14% by training and deploying a BERT-based text classification model for personalized content recommendations, addressing cold-start recommendation challenges for new users
- Decreased model deployment cycle time from 2 weeks to 48 hours by establishing automated CI/CD pipelines using MLflow and Docker for model packaging
- Processed 5M+ daily event logs efficiently by optimizing PySpark ETL data pipelines on AWS EMR, resolving memory bottleneck errors during peak batch processing windows
- Enhanced named entity recognition (NER) accuracy by 19% by fine-tuning RoBERTa on domain-specific customer support tickets, replacing a legacy regex-based extraction system

Skills:

Programming:	Python, SQL, Apache Spark, Kafka
Machine Learning and AI:	Generative AI, AI Agents, LLMs, SLMs, RAG, GraphRAG, Prompt Engineering, NLP, Fine-tuning, PEFT (LoRA/QLoRA)
Frameworks:	PyTorch, TensorFlow, scikit-learn, LangChain, LlamaIndex, Hugging Face Transformers
LLM Infrastructure:	Ollama, vLLM, Pinecone, Milvus, Chroma, pgvector, Retrieval and Reranking Pipelines
MLOps and DevOps:	CI/CD, Docker, Kubernetes, MLflow, Model Versioning, Monitoring and Logging, FastAPI
Cloud and Data:	AWS (EC2, SageMaker, Lambda, S3, ECS, EMR), Azure, PostgreSQL, Redis

Education:

MS in Business Analytics and Artificial Intelligence from The University of Texas at Dallas

Dec 2025

Bachelor's in Business Administration from Guru Gobind Singh Indraprastha University

Nov 2020